Ashesh Kumar

Lead Data Scientist

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Noida, India

Accomplished and Certified Lead Data Scientist with over **13 years** of total IT experience and **7 years** of relevant experience in developing effective Data Science applications, thereby helping companies improve sustainability, reduce overall costs and configure best optimized analytical processes. **Highly proficient with 7 years of experience in** Generative AI, Large Language Models (LLM), Machine Learning, Data Science, Predictive Modeling, Natural Language Processing(NLP), Deep Learning, Python, Big Data, Pyspark, SQL & Azure Services

Professional Experience

Ernst & Young (E&Y) GDS, Noida — May 2021 — Present
Lead Data Scientist

Projects:

UK - R&D Cost Source Prediction Model (Machine learning & Python):

- Designed and executed a **scalable Azure Storage** solution, managing over **10 TB** of data with full compliance and security.
- Developed automated data ingestion and transformation pipelines using **Azure Data Factory**, increasing data processing efficiency by **40%.**
- Created **Supervised Machine learning** models in **python** to predict claim cost source for different clients across domain
- Prepared **Accuracy metrics** to evaluate the model efficiency for cost categories.
- Facilitated successful R&D tax credit claims totaling over **£10 million,** significantly impacting the financial health and research capabilities of client companies.

SQL Query Chatbot Development using OpenAI LLM for UK - R&D Data Access:

- Integrated **OpenAI's LLM** to interpret and translate natural language into **SQL** queries, minimizing the effort & time required to retrieve results for **non-technical** users.
- Developed a **user-centric** chat interface that simplified data access, leading to a **40%** reduction in data retrieval time for end-users.
- Implemented **feedback loops** and performance metrics to refine the chatbot's accuracy, achieving a **82% success rate** in query translation.
- Facilitated **faster insights** from data, supporting more informed and timely decisions in the R&D process.

Deep Learning Model for Patient Enrollment in Clinical Trials (Pyhon, Tensorflow):

- Designed and trained a **deep learning model** using patient **historical data, medical profiles, and trial** criteria to predict enrollment eligibility with high accuracy.
- Conducted **exploratory data analysis (EDA)** to uncover patterns, anomalies, and correlations within the clinical data, informing feature selection and model architecture.
- Achieved a **30% reduction** in patient screening time by accurately predicting enrollment eligibility.
- Lowered operational **costs by 15%** through the reduction of manual screening efforts and improved patient matching.
- Evaluated model with **ROC-AUC, precision-recall curves, and confusion matrices** for model performance assessment.

Accenture, Bengaluru — June 2018 — May 2021
Senior Data Scientist

Projects:

Deep Learning Model for Classification of Banking Consumer Complaints for CFPB:

- Designed a **feed forward neural network** tailored for text classification, utilizing natural language processing (NLP) techniques to interpret and categorize consumer complaints.

- Engineered a robust **preprocessing** pipeline to **clean, tokenize, and vectorize** large volumes of unstructured complaint text data.
- **Automated** the classification of consumer complaints, reducing manual sorting time by **50%.**
- Enabled the **Consumer Financial Protection Bureau** to respond to consumer complaints **40% faster** by streamlining the categorization process.
- Evaluated model with **ROC-AUC, precision-recall curves, and confusion matrices** for model performance assessment.

Unsupervised Anomaly Detection for Electricity Theft Identification(Autoencoders):

- Developed a robust **deep learning** framework capable of learning complex consumption patterns and identifying deviations signaling potential theft.
- Applied advanced machine learning techniques, including **Isolation Forest & autoencoders** to detect anomalies without labeled data.
- Collaborated with utility experts to **incorporate domain knowledge** into the model, enhancing its predictive accuracy and relevance.
- Contributed to a **25% reduction** in electricity theft cases within the first year of model deployment.
- Enabled the utility company to recover significant revenue losses by identifying and addressing fraudulent consumption, resulting in a **15% decrease** in non-technical losses.

WNS Global Services, Bengaluru — June 2017 — June 2018
Data Science Consultant

Projects:

Predictive Analytics for QBE Motor and Liability Claims Adjudication:

- Processed and analyzed over **100,000 historical claim** records to create a clean, reliable dataset for model training.
- Applied **hyperparameter optimization** techniques to improve model performance and reduce the likelihood of overfitting.
- Created a supervised model to predict whether the claim was likely to be **Accepted** or **Denied**
- Prepared **Decile Analysis** to help client identify & devote more time in investigating Denied cases so that financial losses could be prevented by avoiding litigation stage.
- Reduced manual claims sorting efforts by **40%,** allowing adjusters to focus on complex cases.
- Contributed to a 10% decrease in operational costs associated with claims processing.

Predictive Modeling for Rehabilitation Treatment Necessity in QBE Liability Claims:

- Developed a **supervised machine learning model** to classify claimants based on their likelihood of requiring rehabilitation.
- Assisted the medical team in **prioritizing** cases, leading to a more efficient use of internal medical resources.
- Extracted and **selected relevant features** that influence rehabilitation needs, such as injury severity, claimant medical history, and incident particulars.
- Enabled the client to **advise 20%** more claimants for early rehab consultation, improving their recovery prospects.
- Contributed to a reduction in long-term claim costs by facilitating timely rehabilitation, estimated to save the client **15% in associated expenses.**

NTT DATA, Noida — August 2010 — May 2016
Senior Software Engineer

Projects:

Software Projects - (Database & Manual Testing):

- Worked as software testing engineer across domain. The responsiblities comprised testing front & backend applications across various client engagement.

Education

B.E. (Chemical Engineering)
Manipal Instt of Tech, Manipal

Big Data & Analytics (Full Time Certification)
S P Jain School of Global Management, Mumbai

Areas of Expertise

- Data Science/Machine Learning/Predictive Modelling/Deep Learning/NLP/Statistics
- Genarative AI/LLM(Open AI, LLAMA, HuggingFace, Gemini)/RAG systems/Text Analytics
- Pandas/Numpy/Scipy/Sklearn/Tensorflow/Keras
- Python/SQL/Pyspark
- AWS/Kafka/Spark Streaming/Lakehouse
- Azure Cloud - ADLS Gen2, Databricks, Datafactory, Synapse, Unity Catalog, Data Modelling